

The Continuum Hypothesis: Sign-Game or Mathematics?

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Abstract

The Continuum Hypothesis (CH) remains one of the most studied and enigmatic problems in mathematics. Solomon Feferman (2011) argued that it might not be a definite mathematical question. In this paper I revisit Feferman's perspective, developing a more fine-grained notion of mathematical legitimacy that follows in spirit his philosophy of mathematics, but also tries to accommodate it to mathematical practice. This framework allows us however to salvage the core of Feferman's argument, while also revealing that some of his more radical constraints should be weakened. I further argue that, under this refined notion, Feferman probably goes too far in defending the status he attributes to CH.

Key words: Continuum Hypothesis, Conceptual Structuralism, mathematical legitimacy, practice of mathematics.

1 Introduction

Posed by Cantor in 1878, the Continuum Hypothesis turned out to be one of the most intricate and controversial problems in mathematics. The work of Gödel (1940) and Cohen (1966) showed it to be independent of the ZFC axioms of set theory, and therefore impossible to decide using the tools usually available to mathematicians in practice. For this reason, among others, some authors doubt whether it is possible to give a unique and meaningful answer to this mathematical question. One of the most well-known of them is Solomon Feferman, who has defended (2011) that CH is an indefinite and intrinsically vague mathematical problem, due to the fact that it is, according to him, not sufficiently clear to bear a determinate truth value.

In this text I would like to face this problem from a different perspective. In my view, one of the most crucial and delicate features of any meaningful mathematical statement is whether it amounts to more than a mere logical tautology. It is true that almost any mathematical result can be understood as a true statement in a determinate formal system. However, the real meaning and significance of mathematics depend on whether its symbols connect to uses beyond the formal system. In this regard, I subscribe Wittgenstein's following observation:

It is essential to mathematics that its signs are also employed in mufti (...). It is the use outside mathematics, and so the meaning of the signs, that makes the sign-game into mathematics.

Wittgenstein (1978, V, §2)

Wittgenstein uses the phrase “employed in mufti” to mean that mathematical signs are used outside their formal mathematical setting—in ordinary contexts such as science, measurement, engineering, or everyday reasoning. In these contexts, the symbols appear without their *formal mathematical uniform*, that is, outside the strictly formal system.

Several positions in the philosophy of mathematics have struggled to account for this aspect. In particular, it creates a difficulty for formalist and logicist approaches, which seem especially vulnerable to the *paradox of mathematical inference*.¹ Naturalists, such as Quine (1953), argued that mathematical entities and statements must in some sense be continuous with scientific theories. Thus, according to Quine, mathematical objects exist because of them being essential for the formulation of our best scientific theories, a view that came to be known as the *indispensability argument*. However, Maddy (1997) defended that this argument ends up being incompatible with mathematical practice, and thus argued instead in favor of two fundamental complementary maxims: MAXIMIZE and UNIFY, maintaining that practitioners tend to make the mathematical universe as large and rich as possible, provided that consistency is preserved.

Later, I examine the difficulties faced by this latter position. For the moment, however, I focus on another conception of mathematics that also appears resilient to the paradox of mathematical inference: mathematical structuralism, the view to which Feferman was most strongly inclined. On this view, mathematical statements are formulated within structures of mathematical objects, where the relevant features lie in the relations among those objects rather than in the objects themselves. According to most structuralists, the mere existence of a mathematical structure entails an ontological commitment that cannot be reduced to pure logic.² If we begin with any consistent set of axioms, a structure modeling those axioms can be proven to exist. If a structuralist conception of mathematics were to regard all such structures as legitimate mathematical structures, it would also succumb to the inference paradox. In general, however, structuralists do not take this view. This raises a further question: why do certain mathematical structures count as existing, and on what grounds should we accept a logical structure as genuinely mathematical rather than merely formally possible?

I think this last point is crucial to the debate surrounding the Continuum Hypothesis. It is of course true that this statement has a precise and well-defined meaning within the language of set theory. However, the fact that a statement is well defined within a given logical framework does not, in general, entail that it constitutes a meaningful mathematical statement. Otherwise, even the most intricate and irrelevant tautologies would qualify as genuine mathematical claims. If we consider the structures and theories in which CH can be defined to be genuinely mathematical, then further reasons must be given for this—reasons that do not suffer from the problems faced by views such as Quine’s and Maddy’s.

And this is precisely the question this paper seeks to address: whether certain abstract structures, including those involved in the formulation of the Continuum Hypothesis, have progressed beyond what might be regarded, in Wittgenstein’s terms, as a merely formal *sign-game*. In this regard, I agree with Quine that scientific applicability is crucial, and with Maddy on the importance of everyday mathematical practice. However, I do not think either account succeeds in adequately integrating these two criteria. The perspective developed here is also closely related

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²See Shapiro (1997).

to Feferman’s broader view of mathematics, which he described as *conceptual structuralism*, according to which mathematical objects are understood as positions within structures determined by our conceptual practices. On this view, the legitimacy of mathematical statements ultimately depends on whether the structures involved are sufficiently clear and conceptually grounded.

The text is organized as follows. In Section 2, I examine Feferman’s conceptual structuralism, focusing on the roles of clarity, definiteness, and applicability, and on how these bear on the status of the Continuum Hypothesis. In Section 3, I develop a refined applicability criterion and introduce a hierarchical account of mathematically legitimate structures. Section 5 discusses the legitimacy of finite-order arithmetic, while Section 6 turns to the case of abstract set theory and the place of CH within it. In Section 4, I compare this framework with related views, particularly those of Maddy and Shapiro. Finally, Section 7 draws together the main results of the paper.

2 Definiteness in conceptual structuralism: clarity and applicability

As explained in the introduction, this text examines the mathematical significance of certain structures. In this section, I explore how this notion is addressed—often indirectly—within conceptual structuralism, a framework that has engaged extensively with CH. In the next section, I turn to how I would address the difficulties this conception raises for this issue.

From the perspective of conceptual structuralism, what can be evaluated is not the legitimacy of isolated facts and objects, but of the full structures or frameworks that are introduced in mathematical reasoning. For example, if the whole ZFC set theory universe is regarded as mathematically meaningful, then so is CH, since it is perfectly well-defined within that framework. In this regard, what Feferman most values, as mentioned in the introduction, is the degree to which such mathematical structures are conceptually clear and well-grounded. This emphasis is evident in the fifth of the ten theses he proposes to articulate conceptual structuralism:

Basic conceptions differ in their degree of clarity. One may speak of what is true in a given conception, but that notion of truth may only be partial. Truth in full is applicable only to completely clear conceptions.

Feferman (2011, p. 15)

Feferman argues that conceptions such as those required to formulate CH—namely, third-order arithmetic and stronger systems—suffer from an intrinsic lack of clarity. This is particularly evident in set theory, where, as he notes (2011, p. 1), “the concept of a set being arbitrary is vague or underdetermined, and there is no way to sharpen it without violating what it is supposed to be about”. For this reason, in his particular framework for understanding mathematics, he concludes that the CH cannot be assigned a definite truth value.

This argument has been subject to substantial criticism, most directly by Peter Koellner (2016 & 2022). Koellner argues that Solomon Feferman’s notion of mathematical clarity is unstable and can be extended in such a way that even statements of arithmetic would be rendered indefinite—an outcome that Feferman himself explicitly rejects. I have examined this debate in Blanco (2026), where I argue that many of Koellner’s criticisms on this point are well

taken. In that work, I also defended that Feferman’s position can be made more resilient by emphasizing an alternative interpretation of his conception—one that is closer to the notion of mathematical content that concerns me here. This interpretation focuses on what I call the *applicability criterion*, namely the idea that mathematical entities exist through their systematic and consistent appearance in world pictures.

Feferman adheres to this criterion with varying degrees of explicitness, though he emphasizes it quite generally. For example, it appears in Thesis 3 of conceptual structuralism (2011, p. 11):

The basic conceptions of mathematics are of certain kinds of relatively simple ideal world-pictures which are not of objects in isolation but of structures, (...). They are communicated and understood prior to any axiomatics, indeed prior to any systematic logical development.

This passage emphasizes that the most basic and primitive mathematical structures are connected to the world we inhabit, in the sense that they help us understand and model it.³ In another text, Feferman argues that the mathematics applicable to real-world physics can be derived from relatively simple systems such as Peano Arithmetic:

I have come to conjecture that practically all scientifically applicable mathematics can be formalized in systems reducible to PA, or, as I have sloganized it in (1993): a little bit goes a long way. I have learned of a couple of cases (...) where it appears one must go beyond the resources of PA; but the physical theories that require such additional strength are rather speculative.

Feferman (1999, p. 15)

In conclusion, Feferman takes it to be important that mathematical structures are connected to real-world physics, at least insofar as they enable us to construct models and representations of it. However, he does not regard this connection as sufficient to establish the reality of certain mathematical conceptions—particularly those that figure in debates about the definiteness of the Continuum Hypothesis:

Whether or not this thesis will hold for all future applications of mathematics to physical science, what this shows is that the argument from the success of physics to the reality of the set-theoretical conception of the real numbers is completely vitiated.

Feferman (2011, p. 25)

Feferman’s main thesis on this topic, as already mentioned, is that the Continuum Hypothesis (CH) is not a definite mathematical problem. In the present work, however, I am concerned with related but distinct notions, such as mathematical content and the question of whether CH can be regarded as mathematically meaningful. Before explaining in detail, in Section 3, how I understand this concept, I first compare these two notions, which, although distinct, are closely interconnected.

For the moment, let us understand a mathematical statement as meaningful insofar as it makes sense to use it outside the formal systems in which it appears, following a broadly Wittgensteinian idea—see the [Introduction](#). Thus, an indefinite statement in a given formal language is not merely one that is undecidable within a particular system, but rather one whose underlying concepts are insufficiently determinate to fix a definite truth value. Such a statement

³See section 2 in Blanco (2026) for further discussion of this interpretation.

should therefore not be expected to be settled as true or false, even in principle. As a result, it is unlikely to play a significant role in ordinary scientific or mathematical practice, except within the formal framework in which it is formulated. Conversely, the practices in which determinate statements are employed play a central role in shaping both their use and the structures to which they belong. For this reason, even though Feferman does not explicitly formulate his position in terms of meaningful mathematical statements or mathematical content, his reflections on CH clearly bear on these notions, particularly insofar as he emphasizes the role of applicability in scientific and everyday practice.

However, a central difficulty with this emphasis is that it remains too restricted to direct applications in science, especially in physics, thereby overlooking—much like the indispensability argument—important aspects of mathematical practice. In particular, there are statements formulated within otherwise well-established frameworks, such as Fermat’s Last Theorem in PA, for which there is no known way of deciding them without appealing to stronger systems, such as finite-order arithmetic.⁴ At the same time, it is crucial that mathematical entities be integrated into scientific practice in as simple and natural a way as possible. Some of the restrictions Feferman defends—such as limiting the treatment of the continuum to predicative, non-set-theoretical frameworks—do not seem to adequately account for this requirement.

In this regard, I propose to distinguish between two related but distinct notions, both connected to the applicability of mathematical structures. First, closely aligned with what Feferman emphasizes, is the notion of *external applicability*, namely the usefulness of mathematical entities and results in modeling the physical world and thus in being directly applicable in science. Second, I introduce what I call *internal fruitfulness*, that is, the role played by mathematical structures within mathematics itself, insofar as they extend, organize, or deepen our understanding of previously established concepts. From this perspective, there are entities and structures that significantly contribute to mathematical understanding without being directly relatable to physical reality. The aim of the following section—and indeed the central aim of this paper—is to develop a notion of mathematical content that accommodates both of these dimensions, while remaining as faithful as possible to mathematical practice.

3 The refined applicability criterion: mathematically legitimate structures

One of the most paradigmatic cases in the history of mathematics of a concept regarded as obscure, metaphysical, or ill-defined is the development of complex numbers from the sixteenth to the nineteenth century. Before presenting my own approach to this controversy, I outline the reasons why complex numbers in particular shifted from being rejected or ignored to becoming one of the most important concepts in mathematics.

Although the idea of negative square roots had already been mentioned before, it was Rafael Bombelli, in his 1572 work *L’Algebra*, the first to systematically use of what we now call complex numbers.⁵ Bombelli was working on the solution of the reduced cubic equation, an equation

⁴In this regard, see McLarty (2020). I also discuss this problem in more detail in Section 5.

⁵Most of the events mentioned here are important facts in history of mathematics, where I am following texts on the topic such as Nahin (2010) and Christen (2018).

of the form $x^3 + px + q = 0$. Earlier, Cardano had found the following formula to solve such equations:

$$x = \sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} \quad (1)$$

Cardano’s difficulty lay in interpreting this formula when negative square roots were involved. In fact, he concluded that the formula was not solvable “in general.” Bombelli decided to step a little bit further, introducing the symbol *pdm* to represent $\sqrt{-1}$, thereby providing a clear precursor to the imaginary unit.⁶ In this way, he was able to obtain new real roots of certain equations of the given form, which Cardano had been unable to.

Progress in this area accelerated in the following years. While Descartes showed little interest in such numbers, referring to them pejoratively as “imaginary,” Leibniz used them in a manner similar to Bombelli, as intermediate steps in solving equations. Euler was probably the first to use them systematically: he introduced the notation *i* for the imaginary unit and employed it in many formulas. Perhaps the most well known is Euler’s formula, which states that for any real phase α , $e^{i\alpha} = \cos \alpha + i \sin \alpha$. Another notable application is the factorization of fourth-degree polynomials with no real roots (such as $x^4 + 1$). Later, Gauss introduced the complex plane, standardized the notation, and firmly established the acceptance of complex numbers, using them in many important results, most notably in the fundamental theorem of algebra. Since then, complex numbers have become one of the cornerstones of mathematics, and the set of complex numbers is now regarded as a mathematical structure in its own right.

What were the reasons for such a shift? In my view, complex numbers constituted the simplest mathematical structure extending the real line while exhibiting three fundamental features, which had traditionally—albeit in different forms throughout history—been regarded as criteria of legitimacy.

First, complex numbers demonstrated significant problem-solving power, as seen in the factorization and solution of higher-degree polynomials. Second, they provided explanatory unification: results such as the fundamental theorem of algebra, or the unification of trigonometric and exponential functions through Euler’s formula, illustrate this point, and many further examples could be given. Finally, they enabled mathematicians to simplify and extend existing formulations to such an extent that new areas of application emerged, through connections with geometry, analysis, algebra, number theory, and beyond.

These features of structures such as the complex numbers are, in my view, what distinguish mathematical reasoning from mere symbol manipulation. For this reason, I propose to incorporate them into a notion of *internal fruitfulness*, organized through a recursively defined hierarchy of mathematical structures. This is not intended as a formal definition within an axiomatic system, but rather as a philosophical model structured as a sequence of classes of mathematical objects. Let us denote this sequence by $S_0, S_1, S_2, \dots$, representing progressively more encompassing structures.

First, we define the class S_0 , consisting of structures whose legitimacy is grounded directly

⁶This stands literally for *più di meno*, which mean “plus or minus” in Italian—see Christen (2018) for more details. He arrived at many of the operational properties of complex numbers, such as $pdm \cdot pdm = -1$.

in their indispensability for modeling the physical world. This class would plausibly include structures such as the natural numbers $\mathbb{N}$, the rational numbers $\mathbb{Q}$, the Euclidean plane for spatial reasoning, and other core notions underlying real analysis and calculus (such as continuous change). Any structure in S_0 can thus be justified through the requirement of direct external applicability.

Having defined S_0 , we now take the recursive step. A structure X , not belonging to $S = S_0 \cup \dots \cup S_n$, is admitted into the class S_{n+1} if it satisfies, in a standard, simple, and natural way, at least one of the following conditions:

1. **Problem-solving power:** X provides solutions to previously existing, well-defined, and relevant problems formulated within structures in S .
2. **Explanatory unification:** X reveals that phenomena arising in structures in S , previously regarded as disparate, are in fact manifestations of a single underlying principle.
3. **Generative capacity:** X gives rise to new and fruitful lines of inquiry within S .

In such cases, I say that X is *internally fruitful* with respect to the previously defined structures. I maintain that a mathematical structure X is legitimate if $X \in S_n$ for some $n \geq 1$.⁷

The requirement that these conditions be satisfied in a standard, simple, and natural way implies that the legitimacy of certain structures may remain an open question. However, this requirement is essential; otherwise, redundant or artificially constructed structures would also qualify as legitimate. If a structure X is deemed legitimate, then any statement formulated within it may be regarded as a *meaningful mathematical statement*.

For example, the fact that $(\mathbb{C}, +, \cdot)$ is an algebraically closed field indicates that it is the simplest structure satisfying these criteria relative to $\mathbb{R}$. Moreover, complex numbers play a crucial role in simplifying proofs across many areas of mathematics. For these reasons, if a given conception of the real line belongs to S_n , then the corresponding structure of complex numbers naturally belongs to S_{n+1} . For instance, a purely geometric conception of the real line—belonging to S_0 —would correspond to a geometric conception of the complex numbers in S_1 . By contrast, the more standard set-theoretic conception of the real line would correspond to a structure of complex numbers belonging to S_2 , as argued in the previous section.

4 Mathematical content in Maddy’s and Shapiro’s approaches

In binding itself to mathematical practice, this kind of naturalism is in danger of being unduly transitory. Even if one takes the proposed naturalistic point of view and mathematical practice as exemplified in set theory for granted, there is a crucial question as to what determines the “mathematical ends” for which the “most effective means” are to be sought. And having chosen the ends, in what sense does effectiveness justify the means? Why is it to be presumed that the

⁷This notion primarily concerns mathematical structures rather than axiom systems. Nevertheless, structures arise as models of axiom systems, just as such systems can axiomatize structures. This correspondence has been extensively studied in model theory. For this reason, I will occasionally refer directly to axiom systems rather than to the structures that model them.

“good” properties of Borel and analytic sets should generalize to all projective sets, given that they don’t hold for all sets?

Feferman (1999, p. 9)

Different conceptions of the continuum.

5 Legitimacy of finite order arithmetic

6 Legitimacy of abstract set theory

7 Conclusion

Acknowledgments

Conflict of interest

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